

SIH 2026 Solution Blueprint

Four problem statements. One engineering-grade plan.
Zero invented facts.

SIH26156 · ULPF / NTRO SIH26076 · Mausam / IMD SIH26087 · NCCT / Cooperation SIH26138 · Green
Fleet / Egreen Quanta

Built from the verified SIH 2026 scrape (sih.gov.in, 2026-10-02) + authoritative external sources.
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SIH 2026 — Solution Blueprint

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SIH 2026 — Solution Blueprint

Four problem statements. One engineering-grade plan. Zero invented facts.

This document is a build-ready blueprint for four SIH 2026 problem statements: every claim about the problem is sourced from the official statement text or an authoritative external source, and everything uncertain is marked **ASSUMPTION** with an explicit verification step. The four problem statements were taken from a verified local scrape of the official SIH portal (sih.gov.in/sih2026PS , scraped 2026-10-02) and cross-checked against the organisations' own sites, public APIs, and standards bodies.

1. Executive Overview

1.1 The four missions

#	PS	Organisation	Title	Slot pressure
1	SIH26156	National Technical Research Organisation (NTRO)	Universal Log Pre-processing Framework (ULPF)	177/500 submitted
2	SIH26076	India Meteorological Department (IMD, MoES)	Personalized homepage for the 'Mausam' mobile application	322/500 submitted
3	SIH26087	National Council for Cooperative Training (NCCT, Ministry of Cooperation)	AI & LMS-enabled cooperative capacity building, ERP & employment ecosystem	98/500 submitted
4	SIH26138	Egreen Quanta	Quantum-inspired fuel prediction & green fleet optimisation	125/500 submitted

Deadline (per official scrape, all four): **5 October 2026**. An independent third-party archive shows "30 September 2026" for SIH26076 — **ASSUMPTION**: treat 30 September as the safe date and re-verify on the official portal before submission.

1.2 The single most important correction in this document

SIH26138 is a **maritime shipping** problem — vessel types, capacities, cruising speeds, alternative marine fuels (LNG, methanol, hydrogen, ammonia), and shore power — not a road-fleet / EV / driver-behaviour problem. Any team that builds a truck-dashboard with EV charging stations has misread the statement. The Green Fleet blueprint in this document is built entirely around IMO regulation and naval architecture.

1.3 How this document is organised

1. A **verified research dossier** per PS (Section 2) — official facts, then a labelled assumption register.
2. One **deep solution blueprint** per PS (Sections 3–6), each with: product identity, problem analysis, what rival teams will build and how this differs, the out-of-the-box mechanism, full architecture (frontend/backend/AI/data/infra), MVP vs demo vs production scope, screens, a timed demo script with a WOW moment, a submission-format PPT plan, quantified KPI discipline, 15 judge questions with answers, and a risk/mitigation table.
3. A **cross-PS technical strategy** (Section 7): shared platform spine, team of six, 7-phase roadmap, build-vs-mock discipline, and a neutral comparison.

1.4 Reading legend

- **VERIFIED** — confirmed from the official PS text, an official organisation site, a live API reference, or a standards body.
- **ASSUMPTION** — plausible but unconfirmed; every one carries a stated verification step.
- **JUDGE NOTE** — a point phrased for direct reuse in the PPT or the demo narration.

2. Verified Research Dossier

2.1 SIH26156 — ULPF (NTRO)

Field	Value
Category / Theme	Software / Blockchain & Cybersecurity
Submission pressure	177 of 500 (323 open)
Scope (verbatim)	A framework that converts any perimeter network device-generated log or event — regardless of source, format, vendor, or technology — into a standardized, lossless , analytics-ready representation for next-generation SIEM and cybersecurity platforms
Hard requirements (a–k)	Preserve raw without loss · parse source attributes · normalize to common taxonomy · traceability raw ↔ normalized · plug-and-play onboarding · unified visibility · SIEM/data-lake integration · AI/ML-ready output · reduced parser effort · air-gapped deployment · container packaging
Contact listed	"Check nciipc.gov.in , helpdesk1@nciipc.gov.in " — i.e. the National Critical Information Infrastructure Protection Centre is the reference point
Deliverables (VERIFIED , explicitly templated)	Source code link · README with setup · architecture document, max 2 pages · demo video, max 2 minutes · technical presentation, max 5 slides

Key standard (**VERIFIED**, from ocsf.io and the OCSF GitHub schema repo): the **Open Cybersecurity Schema Framework (OCSF)** is the vendor-agnostic, JSON-defined core schema for security events — categories, event classes, objects, attribute dictionary, versioned (schema 1.8.x, server 4.5.x at time of writing). Notable: the old `Security Finding [2001]` class is **deprecated** (since v1.1.0) in favour of specific classes (`Detection Finding [2004]`, `Incident Finding [2005]`, etc.). Any solution that ships 2001 events is demonstrably out of date — a small, checkable detail that signals competence.

Prototype data strategy (sensitivity-safe): use **publicly documented log formats only** — Linux audit/syslog samples, web-access logs, documented CEF/LEEF layouts, and public research collections such as the LogHub/LogPai datasets. No operational, classified, or government data is required at any point, and the demo is designed so none is ever needed.

JUDGE NOTE (PS1): "We never ask for your logs. We prove the framework on public data, and we prove losslessness mathematically — by reconstructing the original bytes from the normalized event and hashing them."

2.2 SIH26076 — Mausam personalised homepage (IMD)

Field	Value
Category / Theme	Software / Smart Automation
Submission pressure	322 of 500 (178 open) — the most competitive of the four
Official statement, in full	Eight user personas with the cards each needs (health, fitness, beach, travel, family, agriculture, commuter, events) — quoted verbatim in Section 4. That persona list is the entire official description. There is no published background, scope, dataset, or API list beyond it.

External verification (**VERIFIED**):

- **Mausam is real and official:** package `com.imd.masuam`, developed jointly by ICRISAT's Digital Agriculture & Youth team and IITM under the MoES Monsoon Mission; surfaces observed weather, forecasts, radar, and warnings from `mausam.imd.gov.in`.
- **Damini (lightning) and Meghdoot (farmer crop guidance) features already exist inside Mausam.** Any team that proposes "lightning alerts" or "farmer advice" as its innovation is proposing something that already ships. This is the single most important differentiation constraint on PS2.
- **IMD runs an official, documented API portal** at `api.imd.gov.in` — 28 endpoints covering city forecasts (7-day, with lat/lon), current weather, district/station nowcasts (including lightning-probability and thunderstorm-severity

categories), district/subdivision warnings (heat wave, fog, ground frost among 17 codes), AWS/ARG station observations, district/state rainfall vs normals, river-basin QPF, marine/port/sea-area/coastal bulletins, cyclone track + wind + cone of uncertainty, sunrise/sunset by lat/lon, highway nowcasts, radar, and agromet advisories.

- **The APIs are key-gated** (a call without a key returns HTTP 401, "API key missing"). Accounts are self-registered with CAPTCHA + email confirmation; government organisations must use official mail; organisational terms go through IMD's nodal officer (Dr. Sankar Nath, Sc-E). **Verified operating constraint (IMD API Portal User Guide):** a key is bound to the caller's static public IP, every call needs a user-bound JWT that expires hourly (`expires_in: 3600`), and the portal allows only 2 DEV + 2 PROD keys — so the key must live on a static-IP backend, never inside the app, and usage statistics must be shared with IMD on request. No published rate limits; data licence is arranged per-organisation, not self-serve. A **CAP RSS warning feed is public domain**.
- **IMD Pune publishes 124 years (1901–2024) of daily gridded rainfall at 0.25° resolution** (135×129 grid), citable as Pai et al. 2014 — a legitimate, public climatology baseline for any "normal vs now" comparison the blueprint proposes.

The structural gap every team must face: the personas demand data **IMD does not produce** — AQI, pollen count, traffic, tides, soil moisture, packing advice. A serious solution therefore cannot be "more tiles." It must be a **decision and provenance layer** that derives advisories from what IMD actually publishes, marks every non-IMD input with its source, and degrades gracefully when a source is absent. That honest constraint is the core of the Section 4 design.

2.3 SIH26087 — Cooperative training, ERP & employment (NCCT)

Field	Value
Category / Theme	Listed as Hardware in the portal, but the statement itself says "Proposed Mode: Software + Hardware " — plan for both, and clarify the hardware (QR/attendance capture device) early in the PPT
Submission pressure	98 of 500 (402 open) — the least crowded of the four
Organisation	NCCT, an autonomous society under the Ministry of Cooperation; network of 20 institutes : VAMNICOM Pune (national), 5 RICMs (Chandigarh, Bengaluru, Kalyani, Gandhinagar, Patna), 14 ICMs, plus ~109 junior training centres academically supported
Scale evidence (VERIFIED , NCCT annual report 2023-24)	3,759 programmes, 227,177 participants trained in one year; VAMNICOM alone exceeded its 10,000-participant target by 185%. The "existing system" is a vast, working, human institution — the proposal must augment it, not replace it. Motto: " Sahakar se Samriddhi " (prosperity through cooperation)
Required feature set (verbatim from PS)	Online registration/nomination · participant/institution/trainee profiles · multilingual e-learning · digital attendance (face/QR) · timetable/hostel/logistics · LMS + assessments + certification · certification repository + verification · career counselling chatbot · employer dashboard · mobile-friendly + offline learning · central database for outreach/monitoring

The strategic reading: the PS lists **eleven features**. Every rival team will build eleven shallow tabs. The winning move is to pick the pipeline the features imply — nomination → training → assessment → **verified skill** → match → hire → outcome feedback — and make the *arrow between assessment and match* the product. Everything else becomes competent-but-ordinary plumbing around one genuinely intelligent core.

2.4 SIH26138 — Quantum-inspired green fleet optimisation (Egreen Quanta)

Field	Value
Category / Theme	Software / Clean & Green Technology
Submission pressure	125 of 500 (375 open)

Field	Value
Organisation (VERIFIED)	Egreen Quanta is a real quantum-technology startup (founder-director Dr. Kumar Gautam). In July 2026 it signed a Strategic Quantum Technology for Port Innovation Agreement with the Mormugao Port Authority — quantum vessel scheduling, cargo-handling optimisation, AI analytics — with MPA providing non-sensitive operational data and pilot support under the Sagarmala / PM Gati Shakti / Green Shipping missions. The company publicly frames future ports as observation → prediction → optimisation → simulation → action loops around digital twins
The five objectives (paraphrased from PS)	Quantum-inspired fuel-consumption prediction across vessel types/conditions · quantum metaheuristic fleet-mix/speed optimisation · minimise fuel, cost, lifecycle GHG · satisfy cargo demand, reliability, emission rules · benchmark against conventional methods on accuracy, convergence, quality, scalability

Regulatory ground truth (**VERIFIED**, IMO):

- **CII (Carbon Intensity Indicator)** is mandatory since 1 Jan 2023 for ships ≥5,000 GT: attained CII = CO₂ mass ÷ transport work, rated **A–E**, with corrective-action plans required for E (1 year) or D (3 consecutive years). The 40%-by-2030-vs-2008 trajectory is the binding context; the CII review and the 2027–2030 reduction factors were due for adoption by 1 Jan 2026 — **ASSUMPTION**: treat factor values as configurable inputs, not constants.
- Ship-level **IMO DCS data is anonymised and aggregate; individual ships are not identifiable**. There is no public ship-level training dataset. Any team claiming "we trained on real fleet data" without a data agreement should be assumed to be bluffing. The honest, winning answer is a **physics-calibrated simulator** documented as such.
- **MEPC.385(81) enhanced granularity** (fuel per consumer type, in-port vs underway split, shore power supplied, laden distance) applies from 2026 reporting — the exact variables a serious model needs, which makes a physics-plus-residual architecture uniquely well-timed.

Critical anti-hallucination boundary (PS4)

No real quantum computer is used or needed. "Quantum-inspired" means classical metaheuristics that borrow quantum concepts (superposition-inspired population states, rotation-gate updates), and the blueprint benchmarks them against an exact MILP baseline on small instances — because a team that cannot beat a MILP on 6 ships has no business claiming quantum advantage.

2.5 Assumption register (all four)

#	Assumption	Verify by
A1	SIH26087 hardware expectation is satisfied by a QR/attendance capture flow on commodity phones, not custom electronics	Ask the SPOC / read the "Delivery Table" template if one is published
A2	IMD API keys are obtainable by a student team within weeks, and a static public IP can be provisioned (self-registration, not gov email)	Register on api.imd.gov.in and provision a static-IP host on day one — the key is IP-bound and every call needs an hourly JWT; if blocked, run the mock on IMD's public GeoServer WFS + cached CAP RSS + gridded climatology
A3	No operational access to MPA/Egreen data exists for the hackathon window	Write to the PS contact early; design the simulator-first architecture so the answer works either way
A4	OCSF 1.8.x attribute/class surface used in code	Pin the vendored <code>dictionary.json</code> + <code>categories.json</code> in the repo at build time
A5	Judging constraints: 5-slide PPT and 2-minute video for ULPF are explicit; the SIH idea submission is a separate artifact — the mandated SIH template, max 6 slides including the title , in PDF	Use the official SIH2026 idea template for the 5 Oct submission; keep the PS-specific caps (e.g. ULPF's 5 slides) for the finale deck

3. PS1 — SIH26156 — LOGSMITH: the Lossless Log Fabric

One-line pitch: A vendor-agnostic log pipeline that converts any perimeter-device log into OCSF events, and *proves* — byte for byte — that nothing was lost.

3.1 Problem, and why the obvious answer fails

SOC teams don't suffer from a shortage of parsers. They suffer from three specific, expensive pains: (1) every new device needs a hand-written parser (~days of regex work) before a single alert can fire; (2) parser outputs silently drop or mangle fields, and nobody notices until an incident review; (3) schemas drift — a vendor firmware update renames a field and downstream detections quietly go blind. Generic answers ("a universal parser with regex + Grok patterns") address none of these verifiably.

3.2 What other teams will probably build

- A Grok/regex pipeline (Logstash-style) that maps Syslog + CEF + a few JSON formats to a flat schema.
- "AI log parsing" that calls a cloud LLM per log line — undemonstrable in an air gap, unscalable, and a privacy non-starter for NTRO.
- A dashboard with pretty charts over parsed logs, where "lossless" is asserted in a bullet point.

3.3 How LOGSMITH is different

Three mechanisms, each directly aimed at a pain above — and each *demonstrably* working in the demo:

1. **The Losslessness Contract.** Every normalized event is wrapped in an envelope containing the original bytes (or a pointer + length), a SHA-256 of the raw event, the mapping version, and a field-level provenance map (which output attribute came from which input span). A standalone `verify` tool **reconstructs the original bytes from the normalized record and compares hashes**. Lossless stops being a claim and becomes a passing test. **This is the product's USP in one sentence.**
2. **Zero-shot source onboarding.** Drop 200–500 sample lines from an unseen device. The induction engine — Drain-style template mining, type clustering, and a *local, CPU-only* field-name → OCSF-attribute proposer constrained to the schema's type system — emits a versioned mapping plus a held-out validation report: per-field precision/recall, coverage %, and a confidence score. A human approves with one click (human-in-the-loop by design).
3. **The Drift & Coverage Radar.** Every mapping continuously reports parse coverage and per-field value-distribution statistics (Population Stability Index vs the validated baseline). When a firmware change renames `src` to `src_ip`, the radar flags the drift event, quarantines affected events under the old mapping version, and routes them for re-induction — detections never silently go blind.

Supporting mechanisms: OCSF-native output with pinned schema version (never the deprecated 2001 class); offline GeoIP/asset enrichment from bundled databases; CEF/LEEF/JSON/XML/key-value/syslog codec library; air-gap packaging as a single OCI image with SBOM and a live "zero external calls" proof; mapping registry with signed versions and regression test vectors.

JUDGE NOTE (PS1): "Every other parser says 'lossless.' We are the only team that can *prove* it, live, in front of you, by rebuilding the original log from our output."

3.4 Core features (mapped to requirements a–k)

Requirement	LOGSMITH answer
(a) preserve raw	Raw bytes + SHA-256 in every envelope; optional external blob store for scale
(b) parse attributes	Drain-template mining + codec library + regex assist for the long tail
(c) normalize taxonomy	Versioned OCSF mappings; <code>metadata.product</code> always set; schema version pinned
(d) traceability	Field-span provenance map + mapping version + content hash per event

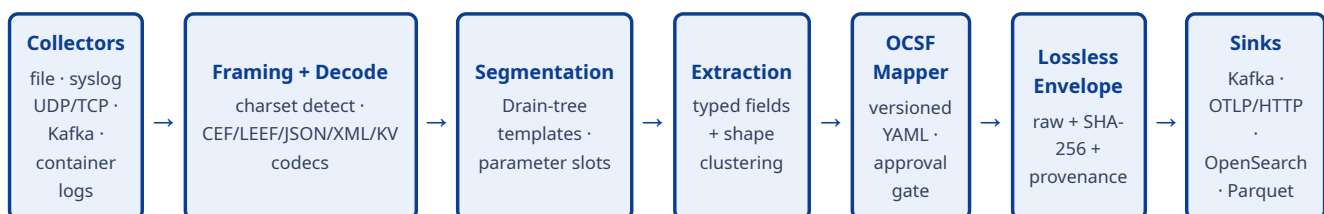
Requirement	LOGSMITH answer
(e) plug-and-play onboarding	Sample-drop induction flow with validation report and one-click approval
(f) unified visibility	Coverage/drift radar per source; unified OCSF event stream for all sinks
(g) SIEM / lake integration	Kafka + HTTP/OTLP + OpenSearch + Parquet-on-S3 sinks; OCSF JSON is the contract
(h) AI/ML-ready	Typed, schema-stable stream + documented residual/anomaly example (Isolation Forest on OCSF features)
(i) reduced parser effort	Measured: mapping induction in minutes with a reported score, vs hand-written rules
(j) air-gapped	Single container, offline assets, SBOM, zero-egress test in CI
(k) container	OCI image; compose file for SIEM-side services

3.5 AI/ML components — with the "why AI?" answered

Component	Model / method	Input → output	Metric	Why ML is necessary; fallback
Template mining	Drain-style fixed-depth parse tree (deterministic algorithm, not a model)	Raw lines → log templates + parameter slots	Templates discovered; tokens stabilised	No fallback needed — offline and deterministic
Field → OCSF proposer	Small local embedding model + cosine over OCSF attribute descriptions + type compatibility rules, CPU-only	Unseen field name + sample values → ranked OCSF attribute candidates	Top-3 accuracy on held-out known mappings	Fallback: pure rule/lexical matching; output is always a <i>proposal</i> a human approves
Type clustering	Shape-feature vectors + HDBSCAN-style clustering	Raw values → inferred semantic types (ip, port, ts, user, path...)	Cluster purity on labelled sample	Fallback: regex type library
Drift detection	PSI / chi-square on per-field distributions (statistics, not ML)	Live stream vs baseline → drift events	Detection latency; false-drift rate	Deterministic, no fallback needed
Demo anomaly layer	Isolation Forest on normalized OCSF numeric/categorical features	OCSF stream → scored anomalies for the "AI-ready output" proof	Precision@k on injected anomalies	Clearly labelled as demonstration of <i>output usability</i> , not a production detector

No cloud LLM anywhere in the pipeline. Say this on slide 1 of the architecture.

3.6 System architecture



Side-plane (control): **Mapping Registry** (signed, versioned mappings + regression vectors), **Coverage/Drift Radar** service, **Verify tool** (reconstruction + hash check), **Induction UI**. State in PostgreSQL; stream backpressure via Kafka; exactly-once semantics through idempotent `event_uid` derived from the content hash.

Frontend: React + Vite + Tailwind — three views only: Onboarding (drop samples → proposed mapping → validation report → approve), Radar (per-source coverage, drift timeline, quarantined events), Event Inspector (raw bytes, envelope, OCSF JSON, provenance spans, verify button). Restraint here is a feature: fewer screens, deeper proof.

Backend: FastAPI (control plane + induction jobs), Go or Rust optional for the hot parse path (**ASSUMPTION:** keep the hot path in Python with multiprocessing first; port only if the 100k logs/sec claim needs defending — benchmark

honestly and report the number you actually measured). Workers: Celery/RQ or plain asyncio queues. Storage: Postgres (mappings, baselines, provenance index) + filesystem/S3 Parquet for raw blobs at scale.

Data pipeline: sample ingest → charset/line detection → codec decode → template mining → field typing → mapping proposal → held-out validation → human approval → versioned publish → live stream with coverage + PSI drift → quarantine + re-induction loop.

Security architecture: no log content ever leaves the box; content hashes only for dedup; RBAC on registry approvals; signed mapping artifacts; audit log of every mapping change (who approved what, when); SBOM + pinned base images; CI egress test proving zero external calls; secrets via env/file mounts, never in mappings.

Scalability story: stateless parse workers behind Kafka partitions; envelope design keeps reconstruction local (no cross-event joins); Parquet partitioning by source×hour; drift statistics are sketch-based (count-min / t-digest style aggregates) so memory is bounded. Target honestly: report measured single-node EPS and the partition-scaling curve.

3.7 Product name, user journey, screens

LOGSMITH — the Lossless Log Fabric. The analyst journey: (1) a new firewall arrives → drop 300 sample lines; (2) the induction report shows 97.4% coverage, 3 flagged fields, top-3 OCSF proposals each; (3) approve → mapping v1.0.0 signed and live; (4) radar shows coverage holding at 99%+; (5) firmware Tuesday renames a field → drift event fires, affected events quarantined *with raw preserved*; (6) re-induction produces v1.1.0, regression vectors pass, detections resume. The demo compresses this into 4 minutes with a pre-seeded unseen device.

3.8 Implementation roadmap (7 phases) + team split + build-vs-mock

Phase	Output	Owner
P1 data + foundation	Codec library (syslog/CEF/LEEF/JSON/KV), OCSF vendor pin, sample corpus, envelope spec	Data + backend
P2 core backend	Collectors → segmentation → mapper → sinks; registry + Postgres	Backend ×2
P3 induction + radar	Template miner, proposer, validation harness, PSI drift service	AI/ML + backend
P4 frontend	Three views; verify/inspector UX	Frontend
P5 integration	Kafka → OpenSearch → notebook round trip; container build; SBOM	DevOps/integration
P6 testing	Regression vectors per mapping; 2-min video script rehearsal; egress test	All
P7 demo polish	Seeded unseen device; drift-injection script; 5-slide deck	Research/presentation + all

Must actually work: ingest → normalize → envelope → verify → radar. **May be mocked:** SIEM-side correlation rules, multi-node clustering numbers (report single-node + projection, labelled as such).

3.9 Demo script (3–5 min) + WOW + PPT

0:00–0:20 problem (analysts write parsers, parsers lie silently); 0:20–0:40 why regex-logstash fails (drift blindness, lossy drops); 0:40–1:00 LOGSMITH + the contract; 1:00–3:30 live: onboard an unseen device → show validation scores → **WOW: hit VERIFY and watch the original log bytes rebuilt from the normalized event with matching hash — then inject a firmware field-rename and watch the radar catch it live**; 3:30–4:00 local-CPU induction + air-gap proof (network panel: zero calls); 4:00–4:30 impact (parser effort minutes not days; quarantine instead of blindness); 4:30–5:00 path to production (signed mappings, regression CI, Kafka scale-out).

PPT (5 slides, hard limit — VERIFIED requirement): 1. The parser problem + drift blindness. 2. LOGSMITH: contract + induction + radar (one diagram). 3. Live proof (verify + coverage numbers from your own runs). 4. Architecture + air-gap packaging. 5. Deployment path (what's real, what's next, what we need from NTRO — e.g., format samples under NDA-free terms).

3.10 KPIs, risks, judge Q&A (abridged here; full set in appendix table)

KPIs (measured, never invented): induction coverage % + per-field precision/recall on held-out samples; time-to-onboard (minutes); single-node events/sec; verify pass rate (must be 100%); drift detection latency; quarantine correctness.

"Metric to measure during pilot" for anything requiring NTRO's real traffic.

Top risks: (1) induction scores poorly on truly adversarial proprietary formats → mitigation: regex-assist lane + human approval gate, scores reported honestly; (2) throughput claims questioned → mitigation: publish measured EPS on stated hardware, never claim billions/day without a cluster you don't have; (3) OCSF version drift → mitigation: pinned, vendored schema + migration notes; (4) air-gap dependency leakage (Python wheels, GeoIP updates) → mitigation: vendored wheels, bundled DB, CI egress test; (5) scope temptation (building a SIEM) → mitigation: hard boundary — LOGSMITH ends at the normalized stream.

Judge Q&A highlights: *Why not Logstash?* — Logstash parses; it doesn't prove losslessness, induce mappings, or watch drift. *How do you handle encrypted/compressed payloads?* — Envelope carries them opaquely with hashes; parsing applies only to declared plaintext spans. *What if the vendor format is binary/proprietary?* — Codec plugin interface + induction over decoded samples; if it can't be decoded, the system says so with a coverage score of 0 rather than guessing. *Why no LLM?* — Air gap, determinism, cost per event; the proposer is a small local model and every output is human-gated. (12 more in the appendix.)

4. PS2 — SIH26076 — MAUSAM: the Personalised Decision Homepage

One-line pitch: The Mausam homepage, rebuilt as a provenance-first decision layer — it keeps IMD's own name and retains IMD's own data, tells eight different lives what to *do*, and lets a judge ask any number where it came from and when.

Naming note. The official statement asks for a *personalised homepage for the 'Mausam' mobile application*. The product therefore keeps the name **Mausam**; it is not a side app and carries no separate codename. Presenting it as the app's adaptive home screen answers the statement directly and removes any "you built a different app" objection.

4.1 Problem, and why the obvious answer fails

The official statement is eight personas, each needing a different slice of weather intelligence. The trap is reading "personalized homepage" as "eight dashboards of tiles." Four facts make tiles insufficient:

1. The personas need data IMD does not publish (AQI, pollen, traffic, tides, soil moisture, UV) — so personalization is a **data-fusion and honesty problem**, not a layout problem.
2. Damini (lightning) and Meghdoot (crop guidance) already ship inside Mausam — "add lightning alerts and farmer advice" is proposing what exists.
3. This is the most crowded of the four statements (322/500 submitted, 178 open).
4. **The field has already split into two shallow camps**, and both are beatable: tiles-over-a-foreign-model (most rivals silently surface a global model such as Open-Meteo under IMD branding), and generic LLM chat. Neither is IMD-authentic; neither is auditable.

4.2 What other teams will probably build (observed in public SIH26076 repos)

- Persona tabs of weather tiles whose real data source is a global model, not IMD — an IMD judge can spot a non-IMD number.
- A generic LLM chatbot ("Ask Mausam anything") that hallucinates AQI with full confidence.
- A competent offline-first clone with "data unavailable" gap cards. **This is now table stakes, not a differentiator** — at least two public repos already ship offline caches and honest gap cards.

4.3 How Mausam is different

The product is not a dashboard. It is a **decision layer** with two mechanisms and one positioning:

1. **The Provenance Toggle (the hook).** One control reveals, on every number on screen: source endpoint, station ID, issue time, and staleness. This is the visceral, demoable moment — and against a field that rebrands foreign models, it is also the integrity claim.
2. **The Replayable Advisory Engine (the depth).** A transparent rule chain consumes only documented IMD endpoints and emits *actions* with their triggering facts ("Avoid outdoor exertion 12:00–16:00 — heat-wave warning (code 9) + station max 44.6 °C"). Every advisory is deterministic: **same inputs** → **same card**, and a judge can hand-re-derive any card. A `replay` endpoint reproduces each advisory from logged inputs. A narration model may *rephrase* but can never *invent* a number — numbers are bound to the data payload by construction.
3. **IMD-native lineage (the positioning).** Every value is either an IMD product or explicitly labelled non-IMD. No silent substitution of a global model.

Non-IMD needs (AQI via CPCB, tides/marine beyond IMD's bulletins, traffic) enter only through a **labelled source-adapter interface**; where no public source exists (pollen is the honest example), the card shows the gap instead of a fabricated number. That restraint is itself a differentiator a judge remembers — and it is strongest on the Health persona, where three of the headline inputs are not IMD's to give.

JUDGE NOTE (PS2): "Every number on this screen can tell you where it came from, when it was issued, and how old it is — and every advisory can be re-derived by hand. No weather app in India does that, including the current one."

4.4 The eight personas — three deep, five light

Eight personas is too many to build well. **Three are built to depth; five ship as competent labelled cards.** The three are chosen because they are the richest in genuine IMD product — and because Agriculture is the product the rival field visibly ignores.

Persona	IMD-native products (of the 28 endpoints)	Not in IMD → honesty card
Agriculture (deep)	Agromet advisory (28, weekly text → action cards) · rainfall vs normal (5, 8, 16, 17) · river-basin QPF (10) · ground-frost warning (code 14) · AWS/ARG observations (9)	soil moisture → labelled gap
Commuter (deep)	NHAI highway nowcast (21) + highway 5-day warning (22) — the most under-used IMD product · district/station nowcast (4, 7) · fog (code 15) · lightning probability (nowcast Cat 6/11/19)	traffic → labelled adapter
Health (deep)	heat wave / hot day / warm night / cold wave / cold day (codes 9–13) · humidity · AWS temperature + "feels like" (9)	AQI (CPCB adapter), UV (not IMD), pollen (no source) — three gap cards, shown proudly
Fitness	sunrise/sunset (15) · heat alerts · wind · "best running window" from hourly risk	none — fully IMD-derivable
Beach & surf	sea-area + coastal + port + fishermen bulletins (11, 12, 13, 23) · cyclone track/wind/cone (18, 19, 20)	tides/water-temp → labelled or gap
Traveller	saved places + destination district warnings + packing rules from forecast deltas	flight status → out of scope, stated
Parents & families	school-commute window + rain/nowcast alerts + severe warnings	none
Event planners	extended forecast + rain-distribution bands + comfort index <i>with its formula</i>	none

Persona caveat (be honest in the PPT): of the three deep personas, **Health is the least IMD-servable** — three of its headline inputs are not IMD's. That is why Health is retained: it is the three-gap showcase that proves the honesty thesis on one screen. Neither Agriculture nor Commuter may be described as "easy"; they are *IMD-rich*, which is different.

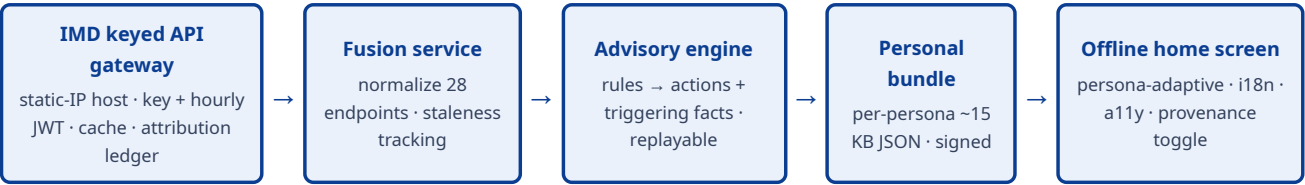
4.5 AI/ML components — with the "why AI?" answered

Component	Method	Why ML / why not
Advisory Engine	Deterministic rule engine over typed IMD facts (the core intelligence)	Rules, because advisories must be auditable and reproducible — a judge can re-derive any card by hand
Narration layer	Small LM, retrieval-grounded, numbers bound to payload	Only for phrasing + translation; falls back to template strings offline
Personal ranking	Lightweight on-device scoring (which cards first for this user at this hour)	No server profile needed — personalization without a user database
Comfort/heat-risk index	Published heat-index + warning-code fusion, formula displayed	Transparent by design; "metric to measure during pilot" for calibration
Climatology delta	Gridded 0.25° normals vs current, per grid cell	Statistics, public data, citable

Explicitly **not** used: forecast generation (IMD does that; we consume it), black-box risk scores, cloud LLMs for facts.

4.6 System architecture

IMD access reality (VERIFIED, from the official IMD API Portal User Guide). The portal at `api.imd.gov.in` exposes 28 documented endpoints, but access is gated: a registered account issues an **API key bound to the caller's static public IP**, and every request needs **both** the `X-API-KEY` header **and** a user-bound `Authorization: Bearer <JWT>` that expires hourly (`expires_in: 3600`). The portal allows **2 DEV + 2 PROD keys**, and states that API usage statistics must be shared with IMD on request. **Consequence:** the key cannot ship inside the app; a small backend on a **static-IP** host must hold it and refresh the JWT. This — not the model — is the project's critical path, and it must start on day one. Until the host exists, the clickable mock draws on IMD's public GeoServer WFS layers so that every displayed value is still genuinely IMD.



Frontend: the Mausam home screen as a lightweight offline-first web app (React + Vite + Tailwind PWA, or a Compose/Dioxus shell) — Leaflet for warning/radar overlays, Workbox service worker, `prefers-reduced-motion` and forced-colors support, language packs for Hindi + 3 more (**ASSUMPTION:** translation quality via human review, never raw MT for warning text). **Backend:** FastAPI fusion/advisory service on a static-IP host; Redis for endpoint caches keyed by issue time; Postgres for saved places + feedback (no behavioural tracking beyond explicit saves); a poller that refreshes the JWT and respects rate etiquette (backoff, conditional fetch). **Data:** IMD keyed APIs; a public **CAP/RSS warning feed** as the always-available floor; gridded climatology shipped as static tiles; labelled CPCB/other adapters. **Infra:** single container + static-hostable PWA; auth only for saved places; a `replay` endpoint so any advisory is reproducible from logged inputs.

4.7 The 5 October deliverable — official 6-slide deck + clickable mock

The idea-submission deadline is **5 October 2026** (extended), and the format is fixed: IMD's mandated template, **a maximum of six slides including the title**, submitted as PDF, points/diagrams rather than paragraphs. So the earlier ten-slide plan is replaced. In the three days before the deadline, ship (a) the six-slide deck and (b) an **offline, clickable mock whose numbers are real captured IMD payloads** — never invented values, or the mock contradicts the one claim being sold. The mock runs from cached/snapshot IMD data and demonstrates the persona-adaptive home screen, the provenance toggle, and one gap card.

Slide	Content
1 · Title	Team, SIH26076 , exact title, MoES / India Meteorological Department, Category: Software, Theme: Smart Automation

Slide	Content
2 · Problem	Eight lives, one app; IMD publishes data, not decisions; current apps show tiles and substitute a global model; scale = 1.4B users
3 · Proposed solution	The persona-adaptive Mausam home screen as a decision layer; one diagram; the provenance toggle is the hook ; replayable advisories is the depth
4 · Technical approach & feasibility	28 IMD endpoints; key + hourly JWT on a static-IP host; deterministic rules engine + <code>replay</code> ; labelled adapters; offline PWA; honest 36-hour build plan
5 · Impact	Agriculture, Commuter, Health built deep; honest gap cards instead of fake numbers; national scalability; qualitative Low/Medium/High — no invented statistics
6 · Research & references	IMD API reference + Portal User Guide; CAP/SACHET; CPCB via data.gov.in; gridded climatology (Pai et al. 2014); public SIH26076 solutions reviewed, and how we differ

4.8 Demo script + WOW

0:00–0:20 problem (one app, eight lives, zero decisions); 0:20–0:40 why tiles + chatbot fail (show a hallucinated AQI answer, then the gap card); 0:40–1:00 the decision layer + provenance principle; 1:00–3:30 live: pick "commuter," 2G-throttle the browser, the home screen loads from cache; a fog morning card appears with an action; **WOW: hit the Provenance Toggle — every number expands to endpoint + station + issue time + age, live**; 3:30–4:00 the advisory engine + climatology delta ("4× drier than the 30-year normal for this grid cell"); 4:00–4:30 impact (comprehension, response time, accessibility — as pilot metrics); 4:30–5:00 deployment (IMD-side host, key management, language expansion).

4.9 KPIs, risks, judge Q&A (highlights)

KPIs: card comprehension (can a user state the action?); time-to-decision; cache-load time at 2G; **advisory reproducibility (target 100% replay)**; translation coverage; per-endpoint staleness SLA (measured); warning-action latency from the CAP feed. No invented percentages.

Top risks: (1) **IMD key/host delay — the account exists but the static-IP host is not yet provisioned, so the keyed API cannot be called today**; mitigation: stand up the host immediately and, until then, run the mock on IMD's public WFS layers so no displayed value is non-IMD; (2) non-IMD sources missing → labelled gap cards, never fake data; (3) "just a UI" perception → the replayable advisory engine is the backend story; (4) language quality → template-first, human-reviewed warning strings; (5) most-crowded PS → provenance + auditability + IMD-native lineage is a defensible moat no tile-app can copy in a week.

Judge Q&A highlights: *Where does AQI come from?* — CPCB adapter, labelled, or absent with explanation. *Why not just use OpenWeatherMap/Open-Meteo?* — Because IMD is the authoritative Indian source and this is an IMD app; mixing sources without lineage is exactly the trust failure we fix. *What if IMD is down?* — Cached bundles + CAP feed + explicit staleness badges; the app degrades to "last known, age shown." *Why rules, not ML, for advisories?* — Auditability: any advisory must be hand-re-derivable in front of a judge. *Is this really a "homepage"?* — It is the app's home screen; the persona is chosen once and the home re-orders. (12 more in the appendix.)

5. PS3 — SIH26087 — SAHAKAR SETU: the Verified Skill-to-Job Ecosystem

One-line pitch: Not a job portal — an evidence pipeline that turns NCCT training into *provable* skills, matches on proof instead of keywords, and learns from every hiring outcome. *Sahakar se Samriddhi, made measurable.*

5.1 Problem, and why the obvious answer fails

NCCT runs one of the largest cooperative training networks in the world — 20 institutes, 227,177 people trained in a single year — and its systems are "largely manual or fragmented." The PS lists eleven features, and the failure mode is obvious: eleven shallow tabs (registration form, video page, QR attendance, job list) that digitise paperwork without changing a single outcome. A trainee still can't answer "what am I missing for the job I want, and exactly what do I do

about it," and an employer still can't trust a certificate they've never seen. The product must make **the arrow between assessment and hiring** intelligent; everything else is plumbing.

5.2 What other teams will probably build

- An ERP clone (registration, timetable, hostel) + embedded YouTube-style videos + a job board with keyword search + a ChatGPT-wrapper "career chatbot" that gives generic advice.
- Face-recognition attendance as the headline innovation — a biometric system on rural youth with no consent, retention, or failure story. A judge *will* ask about this; most teams will have no answer.

5.3 How SAHAKAR SETU is different

One core — the **Competency Graph with Cryptographic Attestation** — plus honest engineering around it:

1. **Skills as evidence, not claims.** Every skill node (mapped to real NCCT programme syllabi) carries: proficiency estimate, confidence interval, and the *evidence artifacts* behind it — item-level assessment responses, module completion with time-on-task, practical submissions, verified attendance. A skill without evidence renders as "unverified" by design. Resumes are inputs; evidence is truth.
2. **Matching as constrained optimisation, not keyword cosine.** For a job posting, the matcher maximises expected fit subject to demonstrated-proficiency thresholds, prerequisite chains from the graph, region/language, and employer must-haves — and every score is *explainable*: the exact missing nodes, each linked to the specific NCCT module (down to institute and duration) that closes it. "What should I do to become job-ready?" gets a concrete answer: *this 3-week module at ICM Nagpur, then re-assess node X.*
3. **The outcome feedback loop.** Employers record hire/no-hire with reasons and (optionally) early performance; the system re-weights skill-importance per role family. The graph learns which credentials actually predict outcomes — the one thing a static portal structurally cannot do. Shown live in the admin view.
4. **Tamper-evident credentials.** Each attestation is a hash-chained record (skill, level, evidence refs, issuer, timestamp) verifiable by any employer without calling NCCT — a public verify page. No blockchain theatre required; hash chains + institutional signatures are sufficient and honestly described.
5. **Offline-first rural reality.** PWA with offline modules, assessments, and QR capture; sync with explicit conflict resolution; Indic-language content; text/light media tiers. Face attendance is **QR-first with coordinator confirmation**; face matching (on-device embedding, explicit consent, deletable, retention policy shown) is an *optional* lane — and the PPT leads with the privacy answer, not the technology.

JUDGE NOTE (PS3): "Every other team matches resumes to job posts. We match *proven abilities* to job requirements — and we can show you the proof behind every single score."

5.4 Module map (the eleven features, competently covered, one core deep)

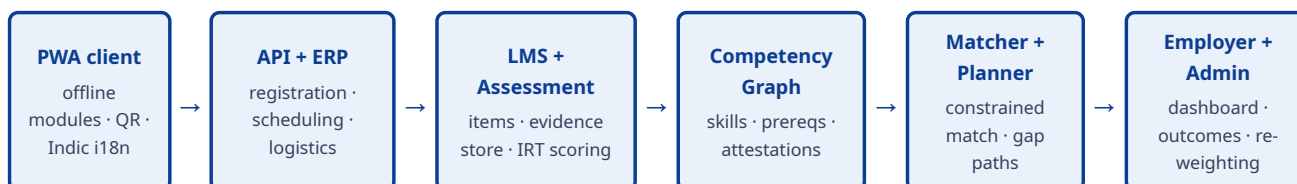
ERP lane: registration/nomination workflows, profiles, timetable/hostel/logistics, central outreach database. Learning lane: multilingual modules, assessments, certification repository + public verification. Employment lane: employer dashboard, postings with structured requirements, explainable matches, outcome capture. Counselling: a **grounded** guidance assistant that answers only from the trainee's own graph + NCCT catalogue ("your gap to Dairy Supervisor is nodes X, Y; the next Y-batch is...") — never generic career advice. Attendance: QR + coordinator confirm as default; face as consented option with on-device processing.

5.5 AI/ML components — with the "why AI?" answered

Component	Method	Why ML / why not
Proficiency estimation	Item-response-style scoring over assessment evidence (per-skill ability + confidence)	Principled uncertainty: the confidence interval is the product, not the point score
Matcher	Constrained optimisation over the competency graph (OR-tools/CP-SAT style)	Optimisation, because requirements are constraints, not vibes; fully explainable by construction

Component	Method	Why ML / why not
Gap planner	Shortest-path over prerequisite edges to target role	Graph algorithms, exact and auditable
Outcome re-weighting	Bayesian/logistic update of skill weights per role family from hire outcomes	The learning loop; labelled "pilot metric" until outcome volume exists
Guidance assistant	Retrieval-grounded generation over trainee graph + catalogue only	Small, constrained, cited; falls back to templates offline

5.6 System architecture



Frontend: React PWA (+ a light USSD/SMS-adjacent notification story via standard gateways — **ASSUMPTION:** gateway access; the core never depends on it). **Backend:** FastAPI or Node; Postgres (operational) + graph represented as relational adjacency (no exotic store needed at this scale); object storage for evidence artifacts; background workers for scoring/sync. **Hardware lane (A1):** commodity Android phones as QR/attendance terminals in a kiosk mode + printed QR cards for trainees — presented as the hardware answer unless the SPOC says otherwise. **Security/privacy:** consent-first biometrics policy, on-device embeddings only, retention + deletion flows, DPDP-act-aligned data minimisation story, role-based access across 20 institutes, audit logs on attestations.

5.7 Demo script + WOW + PPT

0:00–0:20 problem (2.27 lakh trained, outcomes invisible); 0:20–0:40 why portals fail (keyword match demo: same resume, wildly different scores elsewhere); 0:40–1:00 SAHAKAR SETU + evidence principle; 1:00–3:30 live: trainee profile with gaps → click gap → exact NCCT module → **WOW: complete a 5-question live assessment; watch the graph node fill with confidence interval + attestation hash; the same trainee surfaces in the employer dashboard with an explainable score; employer rejects with a reason; the admin panel visibly re-weights the graph** — the full closed loop in ~60 seconds; 3:30–4:00 graph + attestation architecture; 4:00–4:30 impact (pilot metrics: match-to-interview, gap-closure, outcome prediction); 4:30–5:00 rollout across 20 institutes, offline-first.

PPT (≤10): 1. Scale of NCCT + invisible outcomes. 2. Why portals fail. 3. Evidence pipeline (one diagram). 4. Competency graph + attestation. 5. Explainable matching. 6. Outcome loop. 7. Offline + language + attendance honesty. 8. Architecture. 9. Pilot metrics + rollout. 10. Demo + ask.

5.8 KPIs, risks, judge Q&A (highlights)

KPIs: assessment → skill calibration (predicted vs observed task success), match → interview and interview → offer rates, median gap-closure time, employer return rate, offline-sync success, verification-page usage by employers. All "pilot metrics" until measured.

Top risks: (1) biometric backlash → mitigation: QR-first, consent-led, on-device, deletable — lead with this; (2) syllabus mapping effort across institutes → mitigation: seed with 2–3 flagship programmes, show the mapping *method*, not full coverage; (3) cold-start outcomes → mitigation: weights start expert-set, learning labelled as future; (4) offline-sync conflicts → mitigation: explicit, demoed conflict UI; (5) "too big to build" → mitigation: ruthless MVP boundary (Section 7 roadmap phases P1–P4 only).

Judge Q&A highlights: *Why would employers trust your certificates?* — Public verification page + evidence links + institutional signatures; trust is checkable, not claimed. *What stops fake attendance?* — QR nonces + coordinator confirmation + anomaly flags (same device, impossible travel); face only as consented option. *How is this different from SWAYAM/NAPS portals?* — Those deliver learning; we close the loop to *verified hiring outcomes*. *What about trainees without smartphones?* — Kiosk-mode institute devices + printed QR identity + SMS notifications; the trainee never needs a personal smartphone. (12 more in the appendix.)

6. PS4 — SIH26138 — SEAWARD: the CII-Aware Green Fleet Optimizer

One-line pitch: A physics-grounded, quantum-inspired fleet optimiser that predicts fuel honestly, optimises for the regulation that actually bites — the IMO Carbon Intensity Indicator — and shows you the year your "optimal" fleet becomes illegal.

6.1 Problem, and why the obvious answer fails

Maritime fuel is a high-dimensional, non-linear, multi-objective problem: vessel type × capacity × speed × weather × fuel × schedule × regulation. The generic answer — "XGBoost predicts fuel, a genetic algorithm picks ships" — fails three ways: (1) a black-box fuel model cannot extrapolate to a vessel/speed it hasn't seen, which is precisely what fleet planning asks; (2) optimising fuel without the CII rating optimises the wrong objective — a D-rated "efficient" fleet is a compliance failure; (3) "quantum-inspired" without a classical baseline is marketing. Egreen Quanta already talks digital twins and scenario simulation publicly — so a twin alone is table stakes for *this* judge, not a differentiator.

6.2 What other teams will probably build

- An LSTM fuel predictor trained on a CSV of unknown provenance, evaluated with R^2 and no residuals analysis.
- A "quantum genetic algorithm" with no baseline comparison and no statement of what is quantum about it.
- A vessel-tracking dashboard with CO₂ totals — reporting, not optimisation.

6.3 How SEAWARD is different

Four mechanisms, each aimed at a real judge objection:

1. **Physics backbone + ML residuals.** Implement the propulsion power balance honestly: calm-water power from a fitted hull curve (cubic-dominated with documented residual term), plus added-resistance terms for wind/waves/fouling/trim, auxiliary loads, and fuel-specific conversion. Fit a *small* set of physically meaningful parameters per vessel class; gradient boosting learns **only the residuals**. Result: a model that extrapolates to unseen speeds/vessels (physics carries it), degrades gracefully, and is explainable parameter by parameter. **This is the technical heart of the solution.**
2. **CII-native optimisation.** Implement the actual regulation: attained CII = M/W with $M = \sum \text{fuel} \times CF$ and $W = \text{capacity} \times \text{distance}$ (DWT × nm proxy), A–E bands with tightening reduction factors as configurable inputs, E/D-triggered corrective-action logic, and the in-port vs underway + shore-power structure of the enhanced-granularity era. The optimiser maximises fleet utility subject to **rating constraints**, and the benchmark reports rating outcomes, not just tonnes saved.
3. **Honest quantum-inspired optimisation with an exact baseline.** Implement a real quantum-inspired metaheuristic (population with superposition-inspired state encoding and rotation-gate updates — stated plainly as classical code) for the fleet-mix/speed problem, formulated multi-objective (fuel, lifecycle CO₂e, cost, reliability, CII) with a Pareto front — and **solve small instances to optimality with MILP** to show the metaheuristic's gap. The benchmark table the PS demands (accuracy, convergence, quality, scalability) is then credible: conventional = MILP/GA/PSO, proposed = QI-metaheuristic, metrics measured on the same simulator.
4. **The Regulatory-Cliff Scenario Deck.** A timeline slider 2024 → 2035 with tightening CII factors: today's optimal mix visibly slides into D/E territory, and the system re-optimises — newbuild/retrofit, fuel switch (LNG/methanol/ammonia/hydrogen with lifecycle factors), shore-power berths, slow-steaming — showing capex vs compliance per scenario. **This is the WOW moment:** *"Your optimal 2026 fleet is non-compliant by 2030. Here is the cheapest compliant one."*

Data honesty is load-bearing: ship-level DCS data is anonymised and unavailable, so SEAWARD ships with a **documented physics-based voyage simulator** (public engine/fleet particulars, configurable weather, stated distributions) as the training and benchmark ground — and a data-request path to MPA's non-sensitive operational datasets under the July 2026 agreement. The simulator is a feature (reproducible benchmarks), not an embarrassment.

JUDGE NOTE (PS4): "We don't predict fuel with a black box and we don't claim quantum magic. Physics predicts, ML

corrects, MILP verifies, the regulation decides — and we can show you all four on one screen."

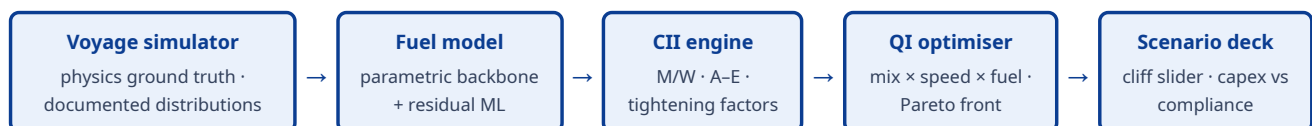
6.4 Objectives coverage (the PS's five, answered)

Prediction → physics+residual model per vessel class/condition. Fleet-mix/speed optimisation → QI-metaheuristic over vessel×capacity×speed with alternative-fuel and shore-power variables. Minimise fuel/cost/lifecycle GHG → Pareto front across all three, lifecycle factors explicit. Demand/reliability/compliance → hard constraints + CII-rating constraints + schedule buffers. Benchmarking → simulator-based harness: MILP (exact, small), GA, PSO vs QI method on accuracy/convergence/quality/scale.

6.5 AI/ML + optimisation components — with the "why" answered

Component	Method	Why this / metric
Fuel model (backbone)	Parametric propulsion balance, fitted per class	Extrapolation + explainability; metric: RMSE + residual bias by speed bin
Residual corrector	Gradient boosting on residuals	Captures fouling/weather/trim effects physics misses; metric: residual RMSE, ablation vs backbone-only
Fleet optimiser	QI population metaheuristic (rotation-gate updates), multi-objective NSGA-style ranking	Global search on non-convex mix problem; metric: hypervolume + gap-to-MILP on small instances
Baselines	MILP (exact), GA, PSO	Credibility; metric: same harness, same seeds
Speed/policy layer	Per-voyage speed optimisation under CII budget	Operationalises the rating; metric: rating attainment per scenario
Shore-power/fuel-switch	Scenario variables with lifecycle CO ₂ e + cost	Decision support; metric: ₹/tonne-CO ₂ e- avoided per scenario

6.6 System architecture



Frontend: React + Leaflet voyage/risk maps (no token), Pareto scatter, Gantt-style schedules, cliff-timeline visualisation.

Backend: Python (NumPy/SciPy) optimisation service + FastAPI; Postgres for fleet/scenario store; job queue for long optimisations with progress streaming. **Data:** simulator-first; public particulars where citable; MPA-request path documented. **Infra:** containerised; deterministic seeds everywhere (every benchmark reproducible); results export for the case-study writeup the PS asks for.

6.7 Demo script + WOW + PPT

0:00–0:20 problem (fuel = cost + emissions + compliance risk); 0:20–0:40 why black-box + dashboard fail (show an LSTM confidently wrong outside its speed range — physics carries ours); 0:40–1:00 SEAWARD + physics → ML → MILP → CII chain; 1:00–3:30 live: load fleet, run optimisation, Pareto front appears, benchmark table fills (QI vs GA/PSO/MILP-gap); **WOW: drag the regulatory-cliff slider to 2030 — the fleet's ratings collapse to D/E, then hit re-optimize and watch the compliant mix (fuel switch + slow-steam + shore power) emerge with capex and ₹/tCO₂e**; 3:30–4:00 CII engine internals (the actual formula, configurable factors); 4:00–4:30 impact (pilot metrics: fuel, rating attainment, cost per tonne avoided); 4:30–5:00 path to MPA pilot (data ask, SEEMP integration, MTM-2027 readiness).

PPT (≤10): 1. The maritime trilemma. 2. Why black boxes fail at sea. 3. Physics backbone + residuals (one diagram). 4. CII-native optimisation. 5. Honest quantum-inspired + MILP proof. 6. Scenario deck + cliff. 7. Benchmarks (measured). 8. Architecture + simulator. 9. Pilot metrics + MPA path. 10. Demo + ask.

6.8 KPIs, risks, judge Q&A (highlights)

KPIs: fuel RMSE by speed bin, residual ablation gain, hypervolume, MILP gap %, rating attainment per scenario, cost per tonne-CO₂e avoided, optimiser wall-clock vs fleet size. All measured on the documented simulator.

Top risks: (1) "quantum" scepticism → mitigation: classical code, stated plainly, MILP-verified; (2) no real data → mitigation: simulator-first is the design, MPA path is the ask; (3) CII factor uncertainty → mitigation: configurable inputs, sensitivity bands shown; (4) physics-model accuracy challenged → mitigation: residual analysis on screen, bias-by-bin reported, never a single R²; (5) scope (lifecycle analysis depth) → mitigation: lifecycle factors as explicit, sourced parameters, not a full LCA claim.

Judge Q&A highlights: *What is quantum about this?* — Nothing runs on quantum hardware; "inspired" refers to specific mechanisms (superposition-encoded populations, rotation-gate updates), benchmarked against classical baselines on identical grounds. *Why not just use MILP?* — We do — for verification at small scale; the metaheuristic is for scale where MILP times out, and we show you the crossover. *Where is your training data?* — There is no public ship-level dataset (DCS is anonymised); our ground truth is a documented simulator, and here are its equations and distributions. *Why optimise CII instead of fuel?* — Because the regulation prices the rating, not the tonne; a fuel-optimal D-rated fleet is a commercial failure. (12 more in the appendix.)

7. Cross-PS Technical Strategy

7.1 The shared platform spine (if one team prepares multiple PS)

All four solutions share a spine: React+Vite+Tailwind PWA shell with a common design system; FastAPI service template with auth/RBAC/audit logging; Postgres + Redis + object storage; Docker Compose → single OCI image; a shared **evidence/provenance library** (content hashes, source ledgers, replay endpoints) that serves ULPF envelopes, Mausam provenance, SAHAKAR attestations, and SEAWARD benchmark seeds. Build the spine once in Phase 1; each PS becomes a different engine on the same chassis. Realistic reuse: ~60–70% of infrastructure code, ~0% of domain engines (they must stay genuinely different).

7.2 Neutral comparison

PS	Solution	Core technology	Main innovation	Demo WOW moment	Main risk	MVP complexity
SIH26156	LOGSMITH	Drain mining + OCSF mapping + PSI drift	Provable losslessness (reconstruct + hash)	VERIFY rebuilds raw bytes live; drift injection caught	Induction quality on adversarial formats	Medium — pipeline + 3 views
SIH26076	Mausam	Rules engine over 28 IMD APIs + offline home screen	Provenance toggle + replayable advisories	Every number reveals source/age live; 2G load	Static-IP key host + crowded field (322/500)	Medium — fusion service + offline screen
SIH26087	SAHAKAR SETU	Competency graph + constrained matching	Evidence skills + outcome learning loop	Live assessment → graph fill → employer match → re-weight	Syllabus-mapping effort; biometric sensitivity	High — ERP+LMS+graph+employer
SIH26138	SEAWARD	Physics fuel model + QI optimiser + CII engine	CII-native optimisation + regulatory-cliff deck	2030 slider breaks the fleet; re-optimize fixes it	No real data; quantum scepticism	High — physics + optimiser + UI

No ranking is offered: pick by team strength (parsers/security → PS1; frontend/PWA + APIs → PS2; full-stack + Hindi content → PS3; Python/math + optimisation → PS4) and by slot pressure (PS3 is the emptiest room; PS2 the fullest).

7.3 Team of six — one structure, four instantiations

Role	Owns	PS1	PS2	PS4 note
Frontend / PWA	Shell, views, offline, i18n, demo UX	Radar + inspector	PWA + toggle	Deck + maps
Backend ×1	APIs, jobs, integrations	Registry + sinks	Fusion + cache	Optimiser service
Data / domain	Pipelines, evaluation, simulator	Codecs + corpus	IMD adapters + climatology	Simulator + physics
AI / optimisation	Models, validation, benchmarks	Induction + drift	Ranking + narration	Residuals + QI + MILP
DevOps / integration	Containers, CI, SBOM, egress tests	Air-gap proof	PWA deploy + poller	Repro harness
Research / presentation	Sources, PPT, video, judge Q&A	OCSF pinning	IMD liaison + key	IMO/MPA liaison

PS3 (largest scope) borrows the second backend headcount into ERP/LMS; PS1/PS2 run lean on five.

7.4 Seven-phase roadmap (all four)

P1 — Data + foundation. Pin sources (OCSF JSON, IMD key + schemas, NCCT syllabi ×3, simulator equations). Verification-first: every A-item in Section 2.5 gets an owner and a date. Output: source ledger + sample corpora. **P2 — Core backend.** The engine's honest core (parse → envelope; fusion → advisory; ERP → evidence; simulator → fuel). Output: API with replay/determinism. **P3 — Intelligence.** Induction/drift; ranking/narration; IRT+matcher; residuals/QI+MILP. Output: measured validation tables (these become PPT slides). **P4 — Frontend.** Only the screens in each blueprint's demo path. Output: clickable demo journey. **P5 — Integration.** Sinks, caches, sync, packaging, SBOM, egress tests. Output: one-command demo environment. **P6 — Testing.** Regression vectors, replay checks, offline/2G passes, video-script rehearsal. Output: green checklist + 2-min video. **P7 — Demo polish.** Seeded scenarios, WOW rehearsal, 15-question drill per PS, slide freeze. Output: submission.

Build-vs-mock discipline: anything the WOW moment touches must be real; anything downstream of the demo (multi-node scale, full syllabus coverage, production auth, MTM-2027 modelling) is explicitly labelled future. Judges punish fake depth more than admitted scope.

7.5 Failure analysis (top risk per PS, with mitigation)

PS	Biggest technical risk	Biggest data risk	Biggest adoption risk	Biggest demo risk	Mitigation posture
PS1	Induction fails on adversarial formats	No NTRO samples available	SOC workflow inertia	Throughput challenged	Approval gates + honest EPS numbers + public-data proof
PS2	Stale-cache wrong advice	Key delays; non-IMD gaps	IMD prefers in-house	"Just a UI" verdict	Replay endpoint + engine-first PPT + A2 fallback
PS3	Graph cold start	Syllabus mapping scale	Institute process change	Scope overwhelm	3-programme seed + ruthless MVP + kiosk hardware story
PS4	Optimiser underperforms MILP at scale	No ship-level data, ever	Pilot data access	"Quantum" dismissed	MILP crossover shown + simulator-first + plain language

7.6 Final recommendations

1. **Register for the IMD API key on day one** (PS2's critical path is administrative, not technical).
2. **Write the two data-request emails in week one** (MPA/Egreen non-sensitive data; NCCT syllabus + SPOC clarification on A1 hardware).
3. **Freeze the WOW moment per PS before writing any other UI** — every other screen serves the 60 seconds that win.
4. **Measure, don't claim:** every PPT number comes from your own validation tables (Sections 3.9/4.8/5.8/6.8).

5. **Respect the explicit deliverable caps** (PS1: 2-page architecture, 2-minute video, 5 slides) — teams lose marks ignoring format, not lacking features.
6. **Say "we don't know" once per presentation**, with the verification step attached. It is the highest-credibility sentence in a hackathon.

Appendix A — Judge Q&A bank (15 per PS, condensed answers)

PS1 — LOGSMITH. 1. Why not Logstash? — Parses, but proves nothing, induces nothing, watches no drift. 2. Encrypted payloads? — Opaque envelope + hash; parse declared spans only. 3. Binary/proprietary formats? — Codec plugins; honest 0% coverage rather than guessing. 4. Why no LLM? — Air gap, determinism, per-event cost; small local proposer; human-gated. 5. Throughput? — Measured single-node EPS on stated hardware + partition curve; no billions/day claims. 6. Which OCSF version? — Pinned/vendored; never deprecated 2001. 7. Schema upgrades? — Migration notes + dual-version replay. 8. False drift storms? — PSI thresholds tuned on validation; quarantine is reversible. 9. Multi-line/stack-trace logs? — Framing stage with session stitching before segmentation. 10. Clock skew? — Provenance records receipt vs event time separately. 11. PII in logs? — Redaction codec lane with audit. 12. Why Kafka? — Backpressure + replay; file mode works without it. 13. Mapping conflicts? — Signed versions; only one active per source; rollback one click. 14. Cost to deploy? — Single container + commodity hardware; pilot metric for scale costs. 15. Copy risk? — Moat is the validation methodology + registry corpus, not the idea.

PS2 — MAUSAM. 1. AQI source? — CPCB adapter or labelled absence. 2. Why not OpenWeatherMap? — IMD is authoritative for India; lineage is the product. 3. IMD down? — Cached bundles + CAP floor + staleness badges. 4. Rules over ML? — Auditability; hand-re-derivable. 5. Hallucination guard? — Numbers bound to payload; LM rephrases only. 6. Languages? — Template-first, reviewed warnings; MT never for alerts. 7. Battery/data cost? — 15 KB bundles, poll-on-change. 8. Accessibility proof? — Screen-reader pass + keyboard map + contrast states in demo. 9. Personal data? — On-device ranking; saved places only, deletable. 10. Packing advice basis? — Forecast deltas + rules shown, e.g. rain PoP thresholds. 11. Meghdoot overlap? — We surface, not duplicate; agri card links advisory provenance. 12. Alert fatigue? — Severity-gated push tiers, user-set. 13. Coastal/marine accuracy? — Verbatim bulletins + cyclone cone; no interpolation claims. 14. Why would IMD adopt? — Adapter, not replacement; attribution + caching reduce their load. 15. Copy risk? — Provenance ledger + offline engine depth.

PS3 — SAHAKAR SETU. 1. Employer trust? — Public verify page + evidence + signatures. 2. Fake attendance? — QR nonces + confirmation + anomaly flags. 3. Biometrics concern? — QR-first; face optional, on-device, consented, deletable. 4. Vs SWAYAM/job portals? — We close assessment → hiring → outcome; they don't. 5. No-smartphone trainees? — Kiosk devices + printed QR + SMS. 6. Syllabus scale? — 3-programme seed + mapping method. 7. Cold-start matching? — Expert weights first; learning labelled future. 8. Gaming assessments? — Item banks + time-on-task + practicals; confidence reflects evidence. 9. Employer adoption? — Structured requirements cheaper than resume triage; pilot metric. 10. Language? — Indic-first content tiers. 11. Certificate fraud? — Hash chains; verification needs no phone call. 12. Rural connectivity? — Offline PWA + explicit sync conflicts. 13. Who governs data? — NCCT institutional ownership; role-based institute scoping. 14. Cost? — Commodity phones + existing infrastructure; pilot metric. 15. Copy risk? — Outcome loop + institute graph corpus.

PS4 — SEAWARD. 1. What's quantum? — Classical code; named mechanisms; no hardware claims. 2. Why not MILP only? — Exact at small scale (we use it); metaheuristic where MILP times out, crossover shown. 3. Training data? — No public ship-level data exists; documented simulator is the ground truth, equations shown. 4. Why CII over fuel? — Regulation prices the rating; fuel-optimal D-rated fleets fail commercially. 5. Physics accuracy? — Residual analysis by speed bin on screen; bias reported. 6. Alternative fuels? — Explicit lifecycle factors + cost; sensitivity bands. 7. Shore power? — Berth-level variable per enhanced-granularity structure. 8. Weather routing? — Added-resistance terms + scenario weather; not a routing solver claim. 9. 2027–2030 factors? — Configurable; review-dated; sensitivity shown. 10. Schedule reliability? — Buffer constraints + Pareto trade-off, not assumed. 11. Egreen's own twin? — We complement: CII-native optimisation + honest benchmarking is the layer twins lack. 12. MPA pilot path? — Data ask + SEEMP-shaped reporting + MTM readiness. 13. Black-box comparison? — Ablation on screen: backbone vs +residuals vs pure ML. 14. Compute cost? — CPU-only; wall-clock vs size curve published. 15. Copy risk? — CII engine correctness + benchmark harness + simulator corpus.

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